



Final Project: Cyber Risk Predictor App

López Gutiérrez Ana Paulina

lopezgutierrezanapaulina20@gmail.com

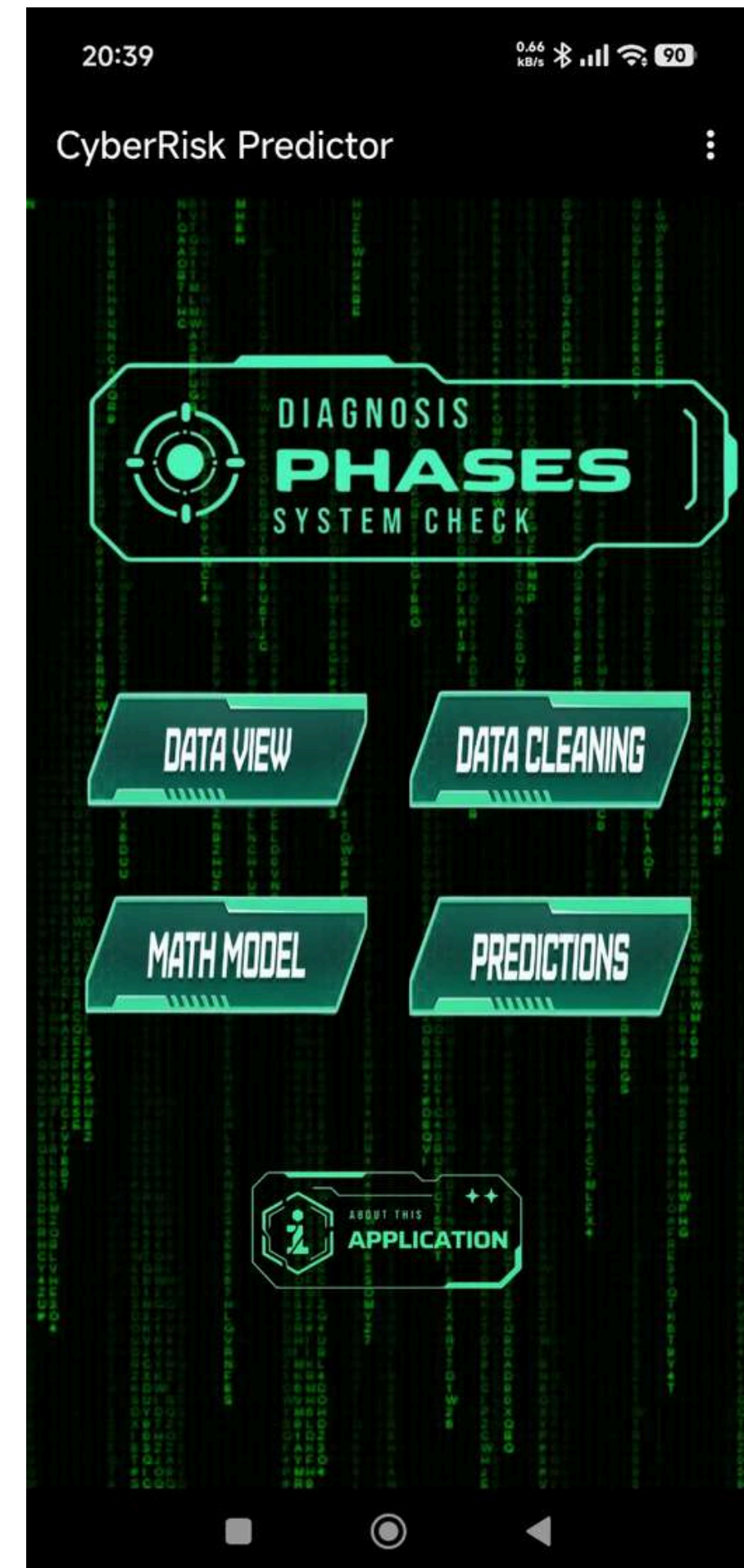
TC PRO B1 G4 2025

DESCRIPTION

What is Cyber Risk App?

CVE Database Explorer, a final project developed in MIT App Inventor, analyzes historical vulnerability records applied to cybersecurity and risk assessment. It implements linear and polynomial regression models to predict the impact of threats (CVSS) based on the attack vector and the complexity level of the exploits.

The database used consists of public standardized cybersecurity records (CVE). All application design and programming are the exclusive property of the Author. All rights reserved.



CyberRisk Database

<https://www.kaggle.com/datasets/thedevastator/exploring-cybersecurity-risk-via-2022-cisa-vulne>

EL DEVASTADOR · ACTUALIZADO HACE 4 AÑOS

< 27 >

Código

Descargar

Riesgo de ciberseguridad (vulnerabilidad CISA 2022)

Gravedad, puntuación CVSS y tipos de vulnerabilidad en seguridad nacional

Tarjeta de datosCódigo (5)Discusión (0)Sugerencias (0)

Acerca del conjunto de datos

Riesgo de ciberseguridad (vulnerabilidad CISA 2022)

Gravedad, puntuación CVSS y tipos de vulnerabilidad en seguridad nacional

Por [fuente](#)

Acerca de este conjunto de datos

Este conjunto de datos presenta una exploración en profundidad de las vulnerabilidades de seguridad en todo Estados Unidos a partir del catálogo CISA Known Exploited Vulnerabilities para 2022. Encontrarás varios tipos de vulnerabilidades y niveles de gravedad, así como puntuaciones CVSS, proyectos de proveedores, nombres de productos y otra información relevante. Al dotarte de este conocimiento, podrás proteger mejor tu red de posibles problemas de seguridad que puedan surgir, ya sea ahora o en el futuro. Profundiza en detalles como vectores de ataque, calificaciones de complejidad y acciones requeridas para cada vulnerabilidad para mantenerte preparado y protegido durante todo tu viaje. Cada detalle importa para mantenerte seguro en línea, así que explora algunos de estos riesgos potenciales y obtén información valiosa hoy mismo.

Ver más

Usabilidad 8.82

Licencia CC0: Dominio Público

Frecuencia esperada de actualización Nunca

Etiquetas

2022-06-08-enriched.csv (296,43 kB)

DetalleCompactoColumna

CVSS	Vector	complejidad	Gravedad				
La puntuación del Sistema Común de Puntuación de Vulnerabilidades asociada a la	El vector asociado a la vulnerabilidad. (Cuerda)	La complejidad asociada a la vulnerabilidad. (Cuerda)	El nivel de gravedad asociado a la vulnerabilidad. (Cuerda)				
9.8 NA Otros (427)	24% 21% 55%	RED NA Otros (159)	58% 21% 21%	BAJO NA Otros (56)	71% 21% 7%	HIGH CRITICA Otros (225)	44% 27% 29%
8.8	RED	BAJO	HIGH				
8.8	RED	BAJO	HIGH				
6.1	RED	BAJO	MEDIUM				
8.8	RED	BAJO	HIGH				
8.8	RED	BAJO	HIGH				
8.8	RED	BAJO	HIGH				
8.8	RED	BAJO	HIGH				
8.8	RED	BAJO	HIGH				
4.3	RED	BAJO	MEDIUM				

Explorador de datos

Versión 2 (1,51 MB)

2022-06-08-enriched.csv

2022-06-09-enriched.csv

2022-06-27-enriched.csv

2022-07-04-enriched.csv

2022-12-09-enriched.csv

Resumen

5 archivos

.csv

80 columnas

https://docs.google.com/spreadsheets/d/14oQXvE62I1oYV8_lyJkDE4iumv3Q5BDf7JwOD0pgb_o/edit?usp=sharing

To ensure optimal application performance, the raw Kaggle dataset underwent a rigorous data cleaning process. Incomplete records, null values, and empty rows were manually filtered out to prevent calculation errors and guarantee maximum accuracy for the predictive mathematical models.

Data Cleaning

CyberRisk Guardado en Drive

Archivo Editar Ver Insertar Formato Datos Herramientas ...

100% Predet... 10

B61 fx ADJACENT_NETWORK Resumir estos datos

	A	B	C	D	E	F	G	H
1	cvss	vector	complexity	severity				
2	9.8	NETWORK	LOW	CRITICAL				
3	7.8	LOCAL	LOW	HIGH				
4	9.8	NETWORK	LOW	CRITICAL				
5	9.8	NETWORK	LOW	CRITICAL				
6	8.8	NETWORK	LOW	HIGH				
7	8.8	NETWORK	LOW	HIGH				
8	9.8	NETWORK	LOW	CRITICAL				
9	9.8	NETWORK	LOW	CRITICAL				
10	9.8	NETWORK	LOW	CRITICAL				
11	8.8	NETWORK	LOW	HIGH				
12	7.8	LOCAL	LOW	HIGH				
13	7.8	LOCAL	LOW	HIGH				
14	7.8	LOCAL	LOW	HIGH				
15	8.1	NETWORK	HIGH	HIGH				
16	9.8	NETWORK	LOW	CRITICAL				
17	7.5	NETWORK	LOW	HIGH				
18	7.8	LOCAL	LOW	HIGH				
19	8.1	NETWORK	HIGH	HIGH				
20	7.5	NETWORK	HIGH	HIGH				
21	9.8	NETWORK	LOW	CRITICAL				
22	10	NETWORK	LOW	CRITICAL				
23	8.1	NETWORK	HIGH	HIGH				
24	8.8	NETWORK	LOW	HIGH				
25	7.5	NETWORK	LOW	HIGH				
26	7.8	LOCAL	LOW	HIGH				
27	7.8	LOCAL	LOW	HIGH				
28	7.8	LOCAL	LOW	HIGH				
29	5.5	LOCAL	LOW	MEDIUM				
30	7.8	LOCAL	LOW	HIGH				
31	8.8	NETWORK	LOW	HIGH				
32	4.3	NETWORK	LOW	MEDIUM				
33	8.8	NETWORK	LOW	HIGH				

To enable mathematical regression within the application, categorical variables were transformed into ordinal numerical values.

Vector (Access Type)

- LOCAL -> 1
- ADJACENT_NETWORK -> 2
- NETWORK -> 3

Complexity (Attack Complexity)

- LOW -> 1
- HIGH -> 2

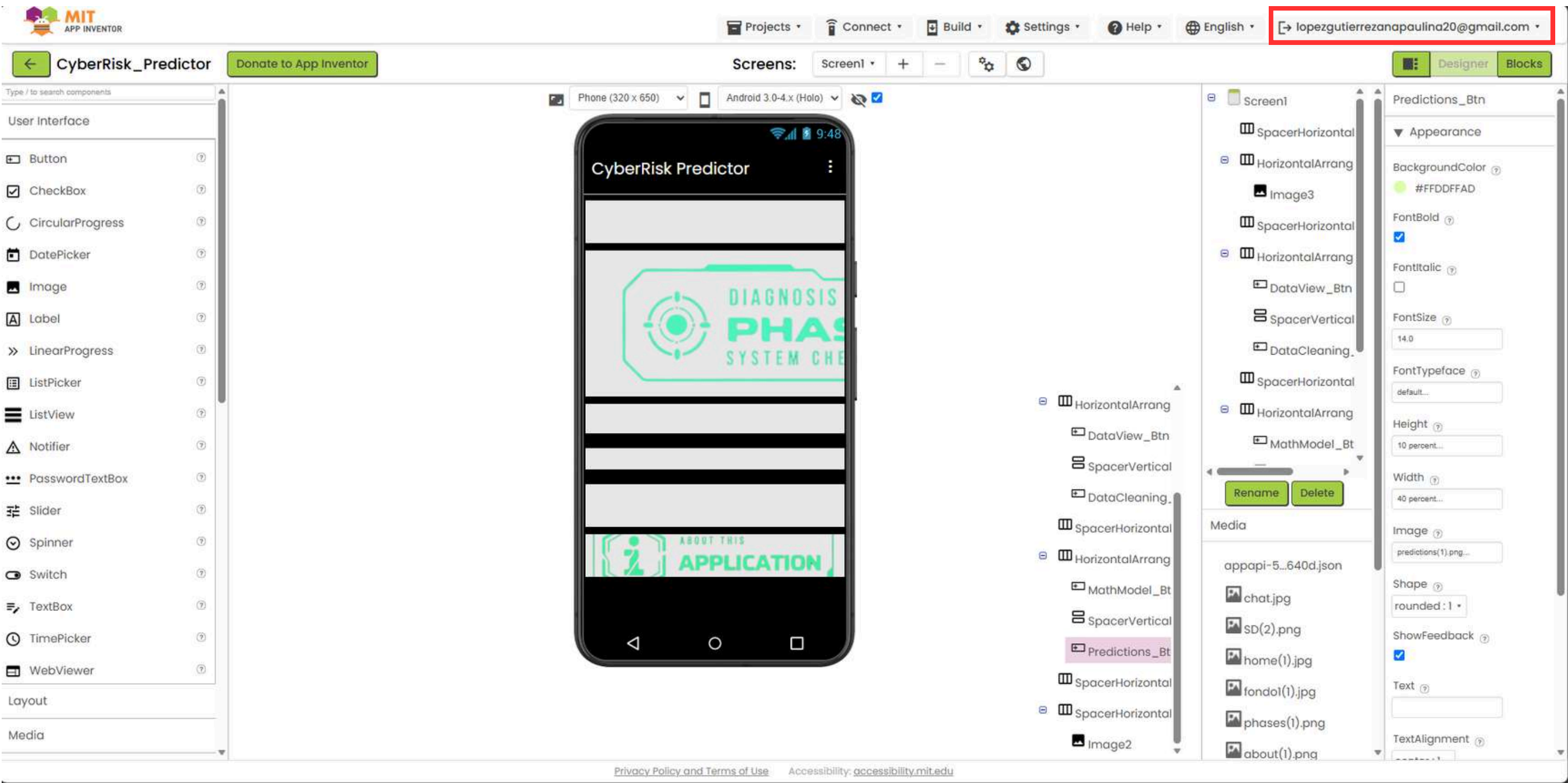
Severity (Severity Level)

- LOW -> 1
- MEDIUM -> 2
- HIGH -> 3
- CRITICAL -> 4

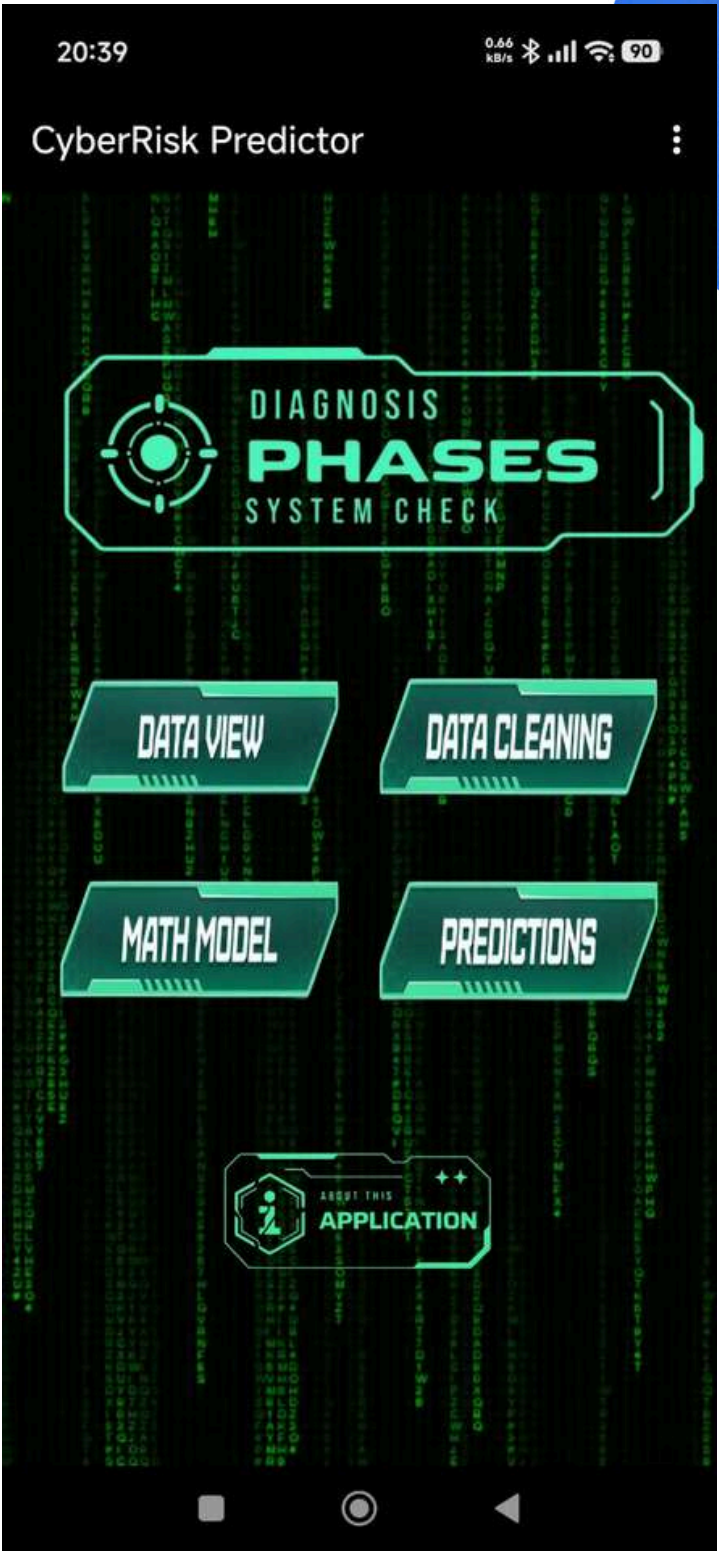


Interface

SCREEN



RESULT



BLOCK PROGRAMMING

MIT APP INVENTOR

Projects Connect Build Settings Help English

lopezgutierrezanapaulina20@gmail.com

CyberRisk_Predictor

Donate to App Inventor

Screens: Screen1 + -

Designer Blocks

Built-in

Control Logic Math Matrices Text Lists Dictionaries Colors Variables Procedures

Screen1

Rename Delete

Media

appapi-5...640d.json chat.jpg SD(2).png home(1).jpg fondo1(1).jpg phases(1).png about(1).png

Show Warnings

when DataView_Btn .Click

do open another screen screenName DataView

when DataCleaning_Btn .Click

do open another screen screenName DataCleaning

when MathModel_Btn .Click

do open another screen screenName Math_Model

when Predictions_Btn .Click

do open another screen screenName makePredictionsScreen

when Image2 .Click

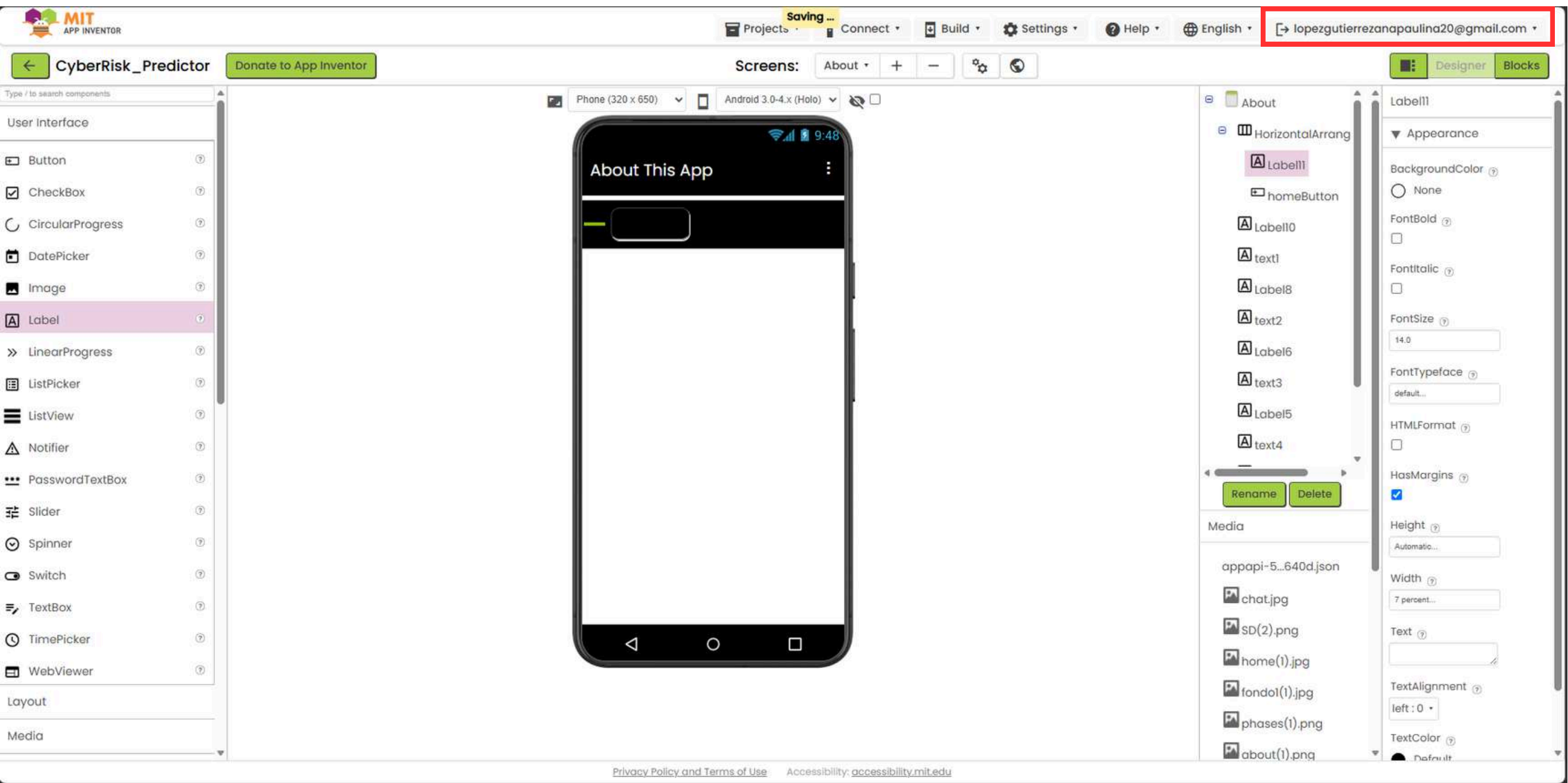
do open another screen screenName About

Privacy Policy and Terms of Use

Accessibility: accessibility.mit.edu

Interface

SCREEN



RESULT



BLOCK PROGRAMMING

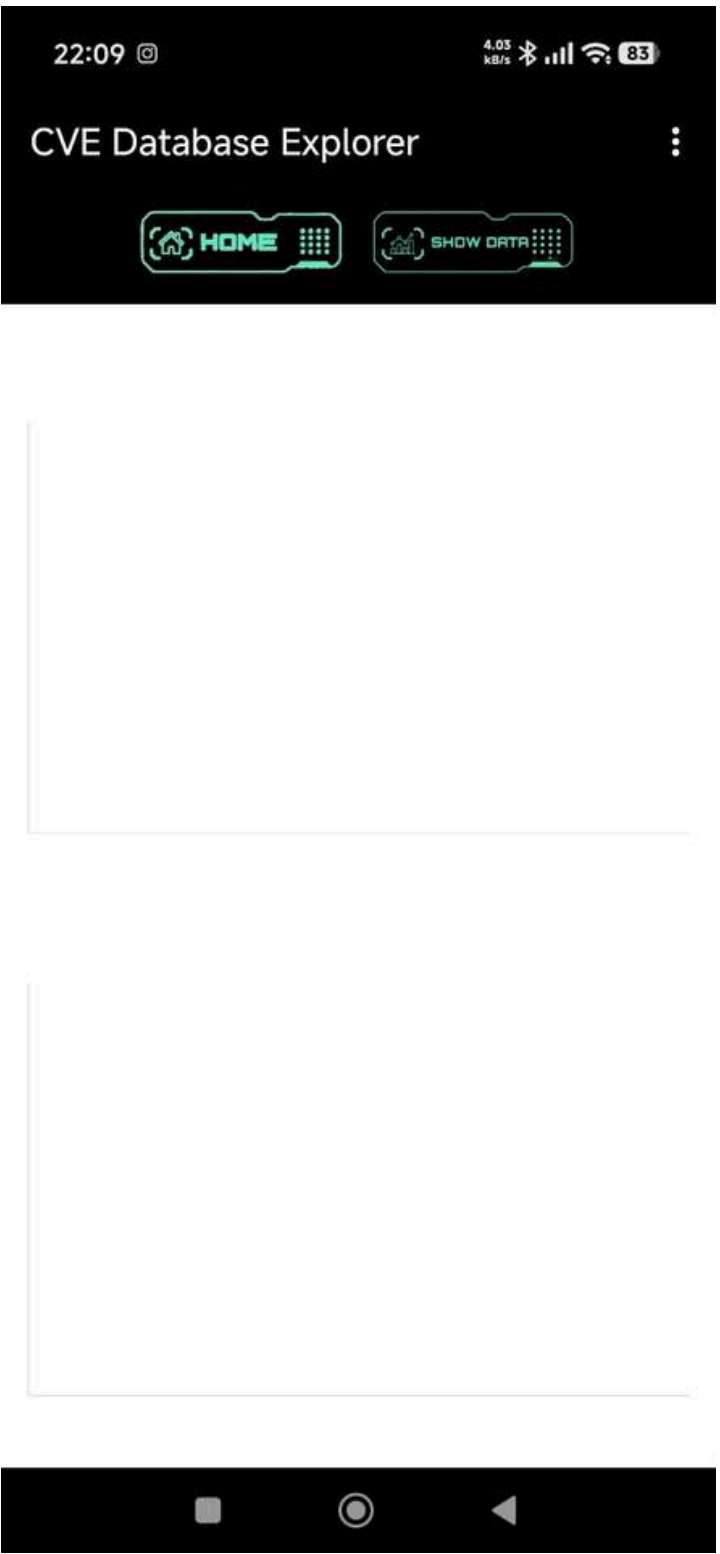
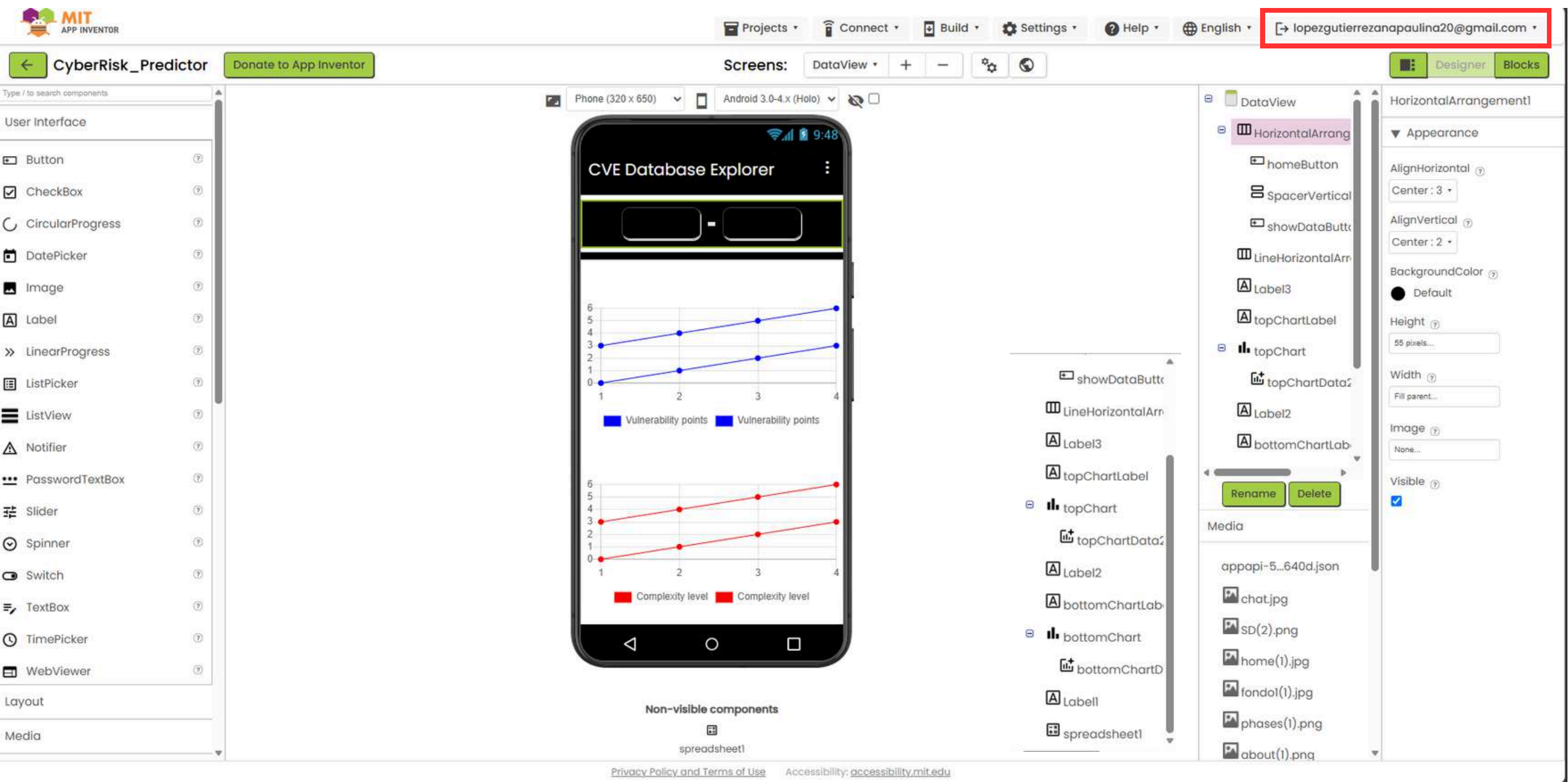
The screenshot displays the MIT App Inventor web interface. At the top, the MIT App Inventor logo is on the left, and navigation links for Projects, Connect, Build, Settings, Help, and English are on the right. A red box highlights the user's email address, `lopezgutierrezanapaulina20@gmail.com`, in the top right corner. Below the navigation bar, the project name `CyberRisk_Predictor` is shown, along with a `Donate to App Inventor` button. The `Screens:` section shows a list of screens: `About`, `+`, `-`, `Settings`, and `Screen1`. The `Designer` tab is selected, showing a visual representation of the app's interface. On the left, the `Built-in` library contains categories like `Control`, `Logic`, `Math`, `Matrices`, `Text`, `Lists`, `Dictionaries`, `Colors`, `Variables`, `Procedures`, and `About`. The `Media` section lists various image files. The main workspace shows a block of code: `when homeButton.Click` followed by `do open another screen screenName Screen1`. A `Show Warnings` button is located at the bottom left of the workspace. The bottom of the interface includes links for `Privacy Policy and Terms of Use` and `Accessibility: accessibility.mit.edu`.



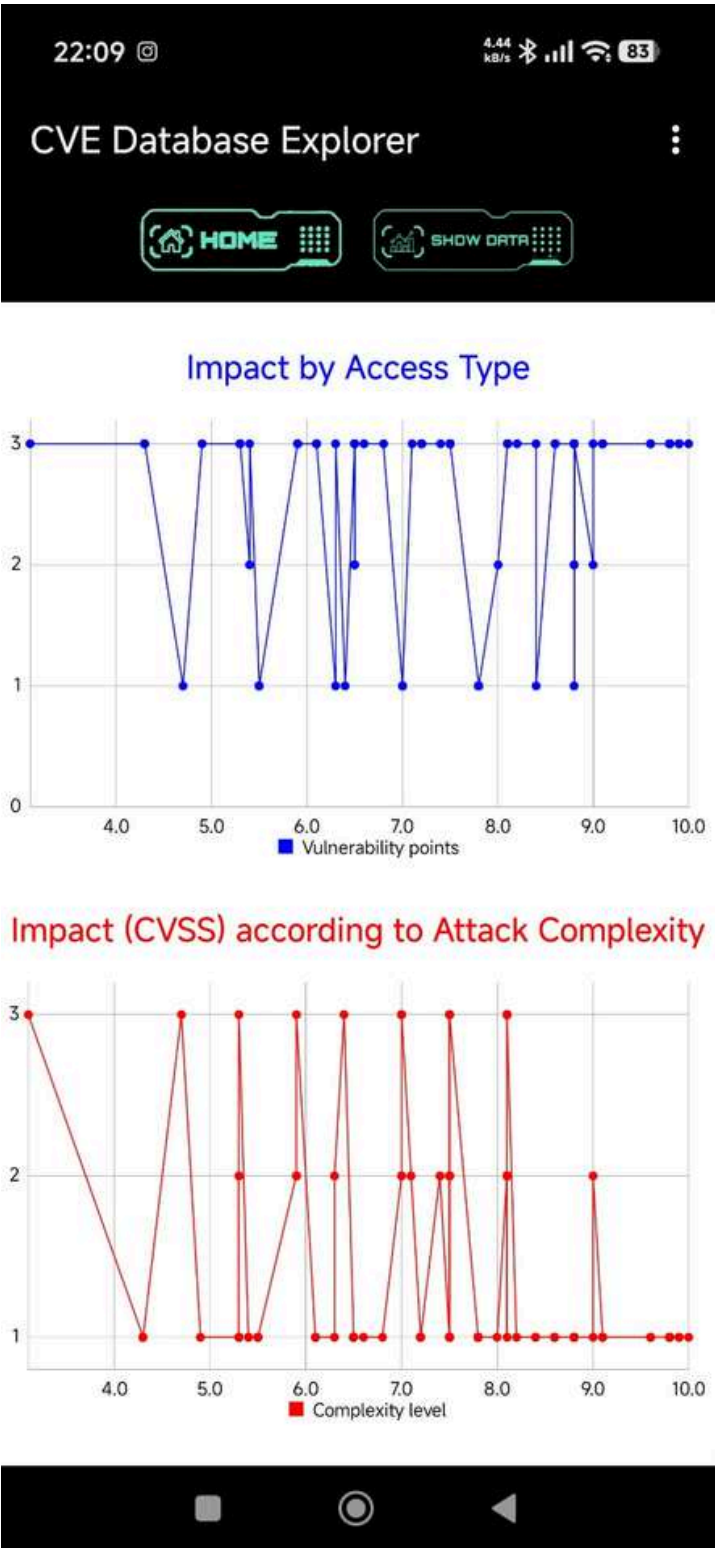
Interface

SCREEN

RESULT



RESULT



BLOCK PROGRAMMING

MIT APP INVENTOR

Projects Connect Build Settings Help English lopezgutierrezanapaulina20@gmail.com

CyberRisk_Predictor Donate to App Inventor

Screens: DataView + -

Built-in

- Control
- Logic
- Math
- Matrices
- Text
- Lists
- Dictionaries
- Colors
- Variables
- Procedures
- DataView

Media

- appapi-5...640d.json
- chat.jpg
- SD(2).png
- home(1).jpg
- fondol(1).jpg
- phases(1).png
- about(1).png

when homeButton Click

do open another screen screenName Screen1

when showDataButton Click

do

- call topChartData2D Clear
- call bottomChartData2D Clear
- call spreadsheet1 ReadSheet sheetName CyberRisk

when spreadsheet1 GotSheetData

do

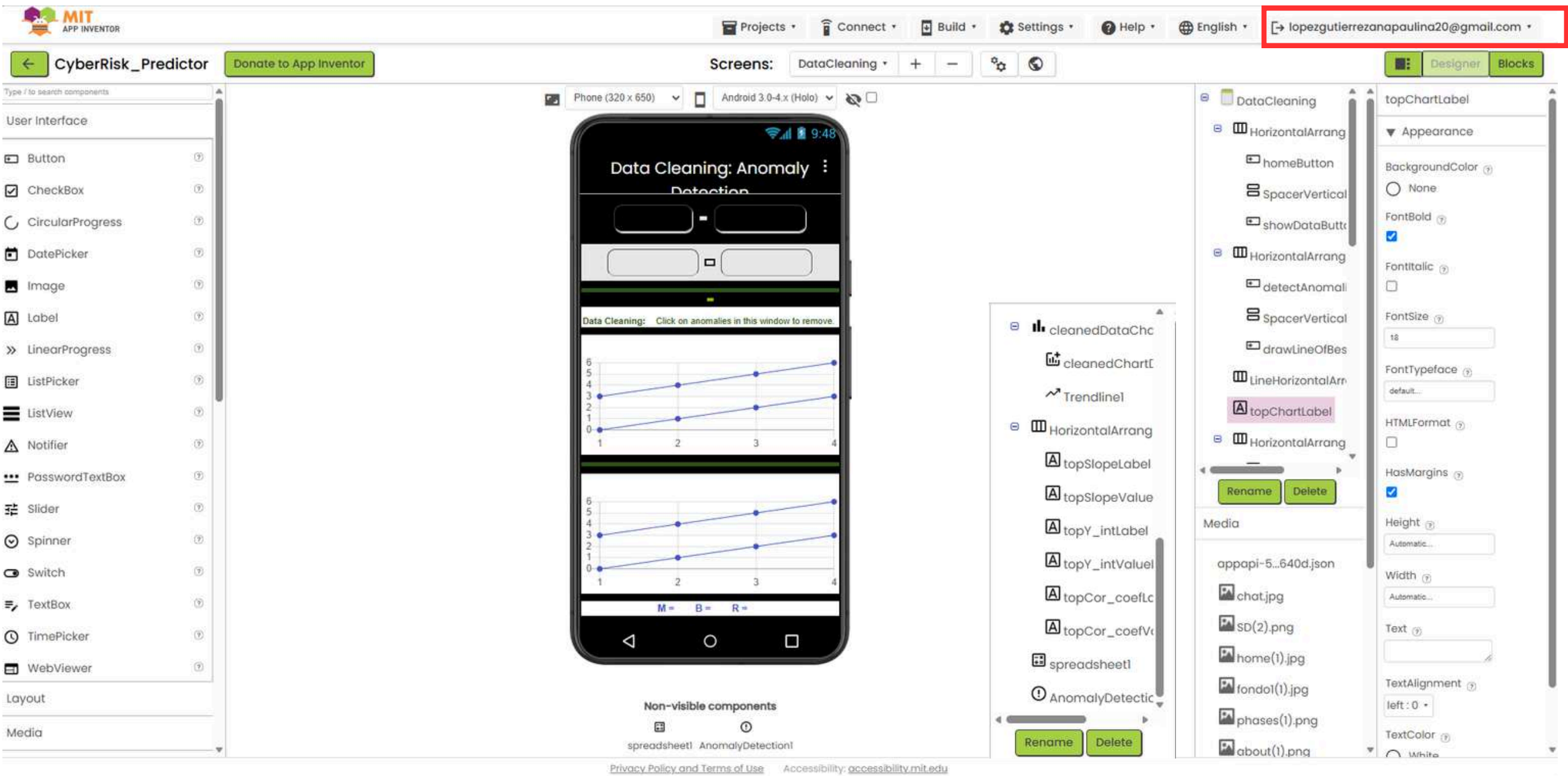
- set topChartLabel Text to Impact by Access Type
- call topChartData2D ImportFromSpreadsheet
- spreadsheet spreadsheet1
- xColumn CVSS
- yColumn vector
- useHeaders true
- set bottomChartLabel Text to Impact (CVSS) according to Attack Complexity
- call bottomChartData2D ImportFromSpreadsheet
- spreadsheet spreadsheet1
- xColumn CVSS
- yColumn complexity
- useHeaders true

Show Warnings

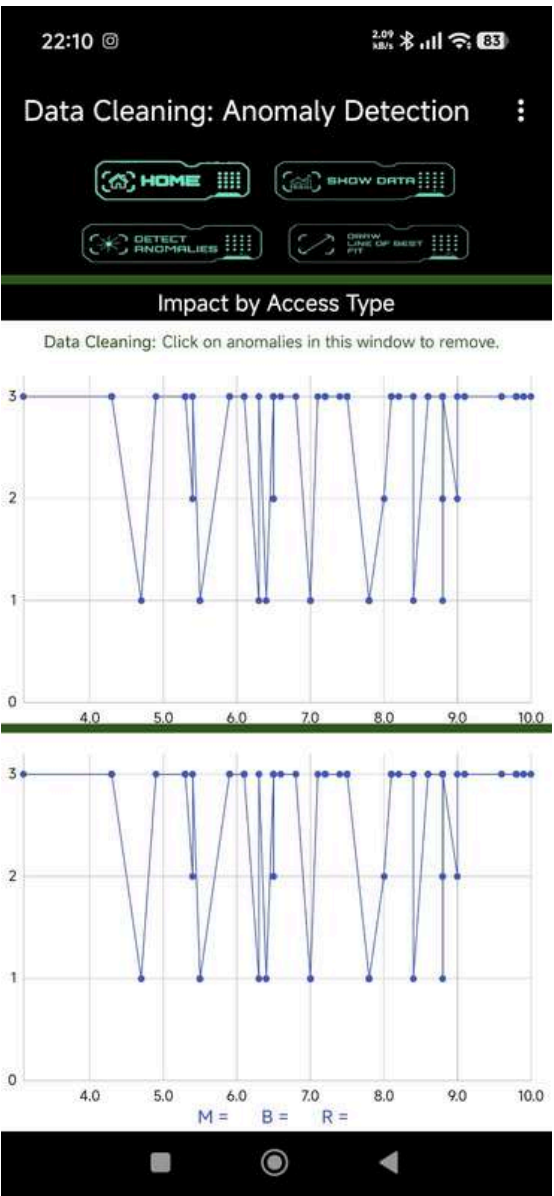
Privacy Policy and Terms of Use Accessibility: accessibility.mit.edu

Interface

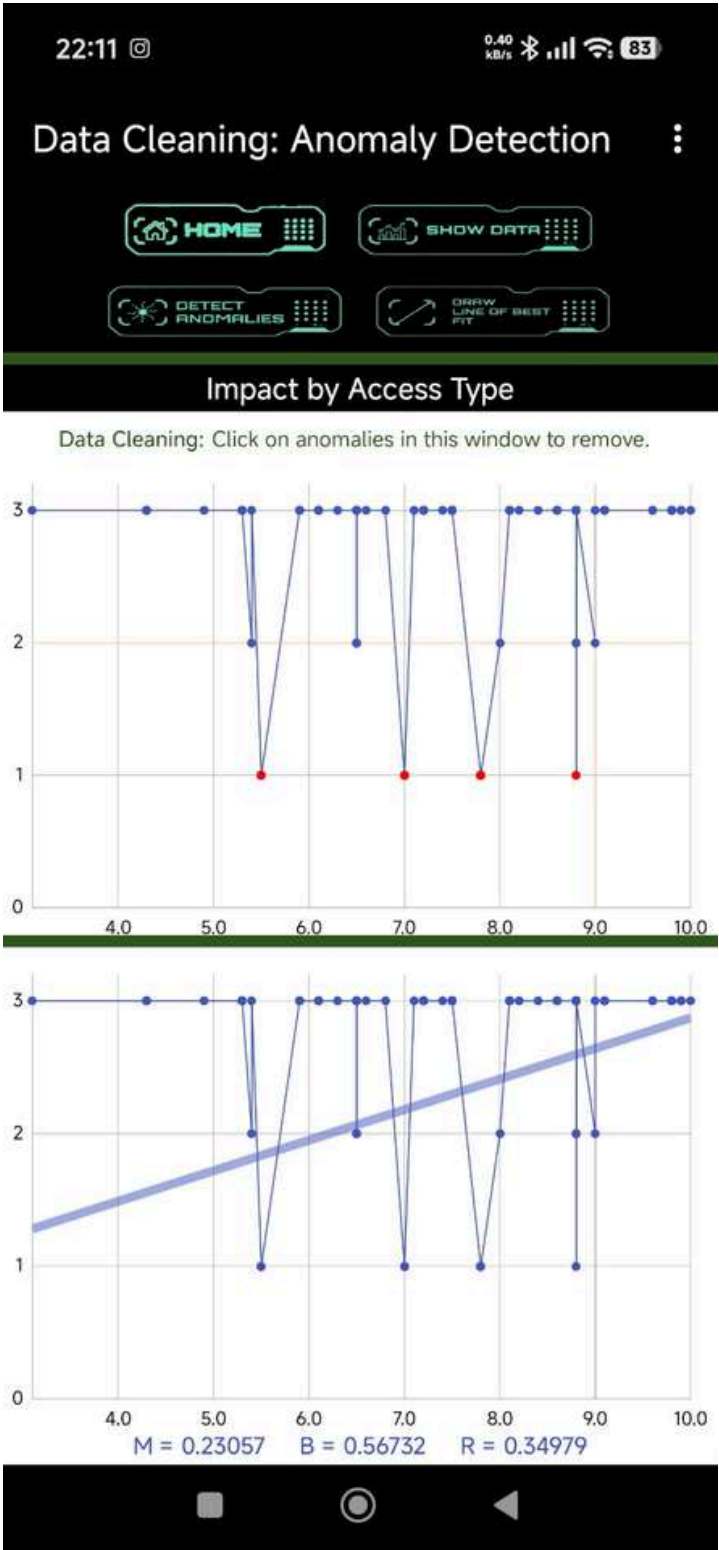
SCREEN



RESULT



RESULT



BLOCK PROGRAMMING

MIT APP INVENTOR

CyberRisk_Predictor Donate to App Inventor

Screens: DataCleaning + -

Designer Blocks

Built-in: Control, Logic, Math, Matrices, Text, Lists, Dictionaries, Colors, Variables, Procedures, DataCleaning

Media: appapi-5...640d.json, chat.jpg, SD(2).png, home(1).jpg, fondol(1).jpg, phases(1).png, about(1).png

when homeButton.Click do open another screen screenName Screen1

when showDataButton.Click do call cleanedChartData2D.Clear call dataCleaningChartData2D.Clear call spreadsheet1.ReadSheet sheetName CyberRisk

when spreadsheet1.GotSheetData do set topChartLabel.Text to Impact by Access Type call cleanedChartData2D.ImportFromSpreadsheet spreadsheet spreadsheet1 xColumn cvss yColumn vector useHeaders true call dataCleaningChartData2D.ImportFromSpreadsheet spreadsheet spreadsheet1 xColumn cvss yColumn vector useHeaders true

when detectAnomaliesButton.Click do call dataCleaningChartData2D.HighlightDataPoints dataPoints call AnomalyDetection1.DetectAnomaliesInChartData chartData dataCleaningChartData2D threshold 1 color red

when dataCleaningChartData2D.EntryClick do if Is in list? thing make a list get X get Y list call AnomalyDetection1.DetectAnomaliesInChartData chartData dataCleaningChartData2D threshold 1 then call dataCleaningChartData2D.RemoveEntry x get X y get Y call cleanedChartData2D.Clear call cleanedChartData2D.ImportFromList list call dataCleaningChartData2D.GetAllEntries

when Trendline1.Updated results do set topSlopeValueLabel.Text to Trendline1.LinearCoefficient set topY_intValueLabel.Text to Trendline1.YIntercept set topCor_coefValueLabel.Text to Trendline1.CorrelationCoefficient

when drawLineOfBestFitButton.Click do set Trendline1.ChartData to cleanedChartData2D

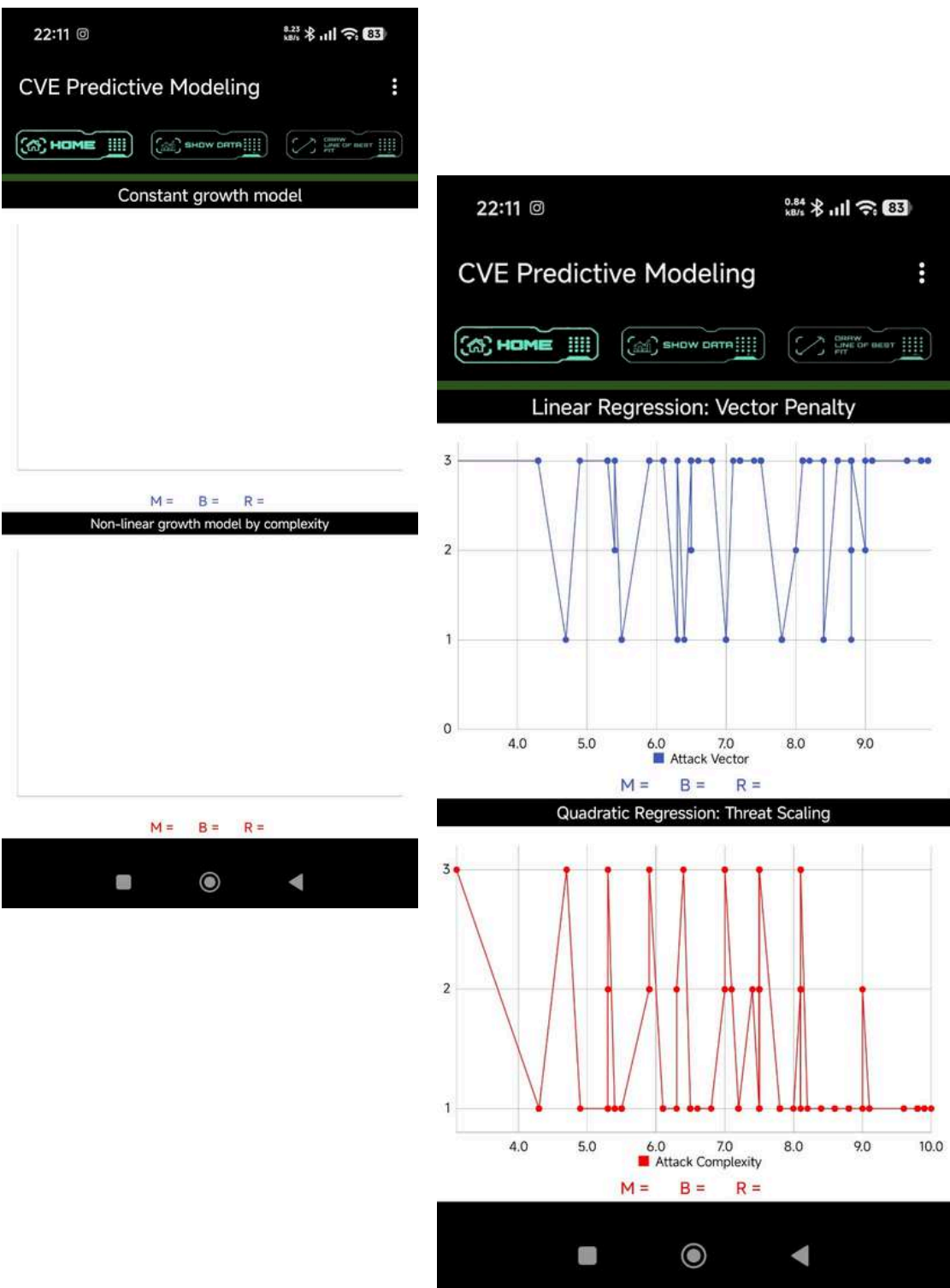
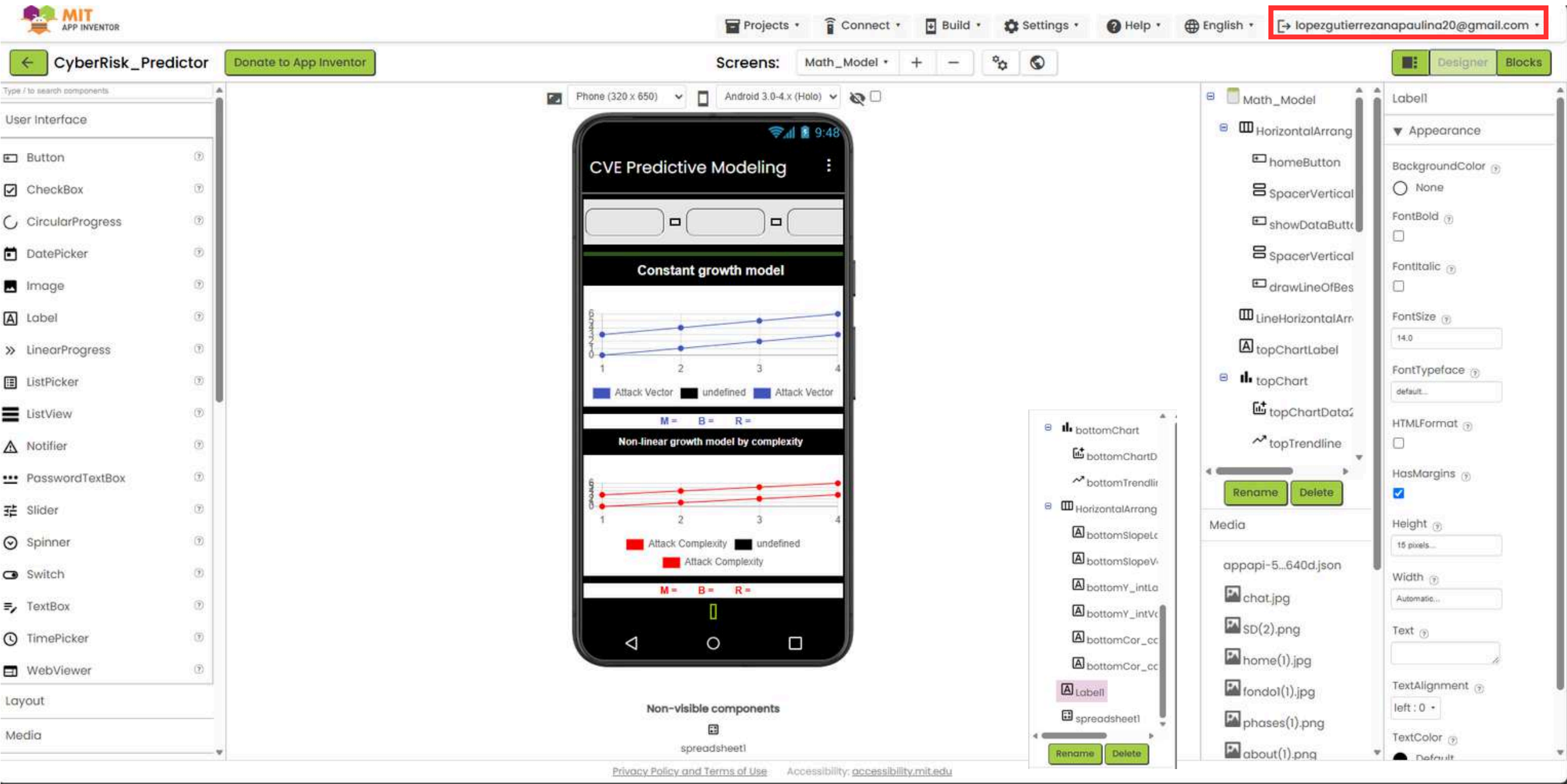
Privacy Policy and Terms of Use Accessibility: accessibility.mit.edu

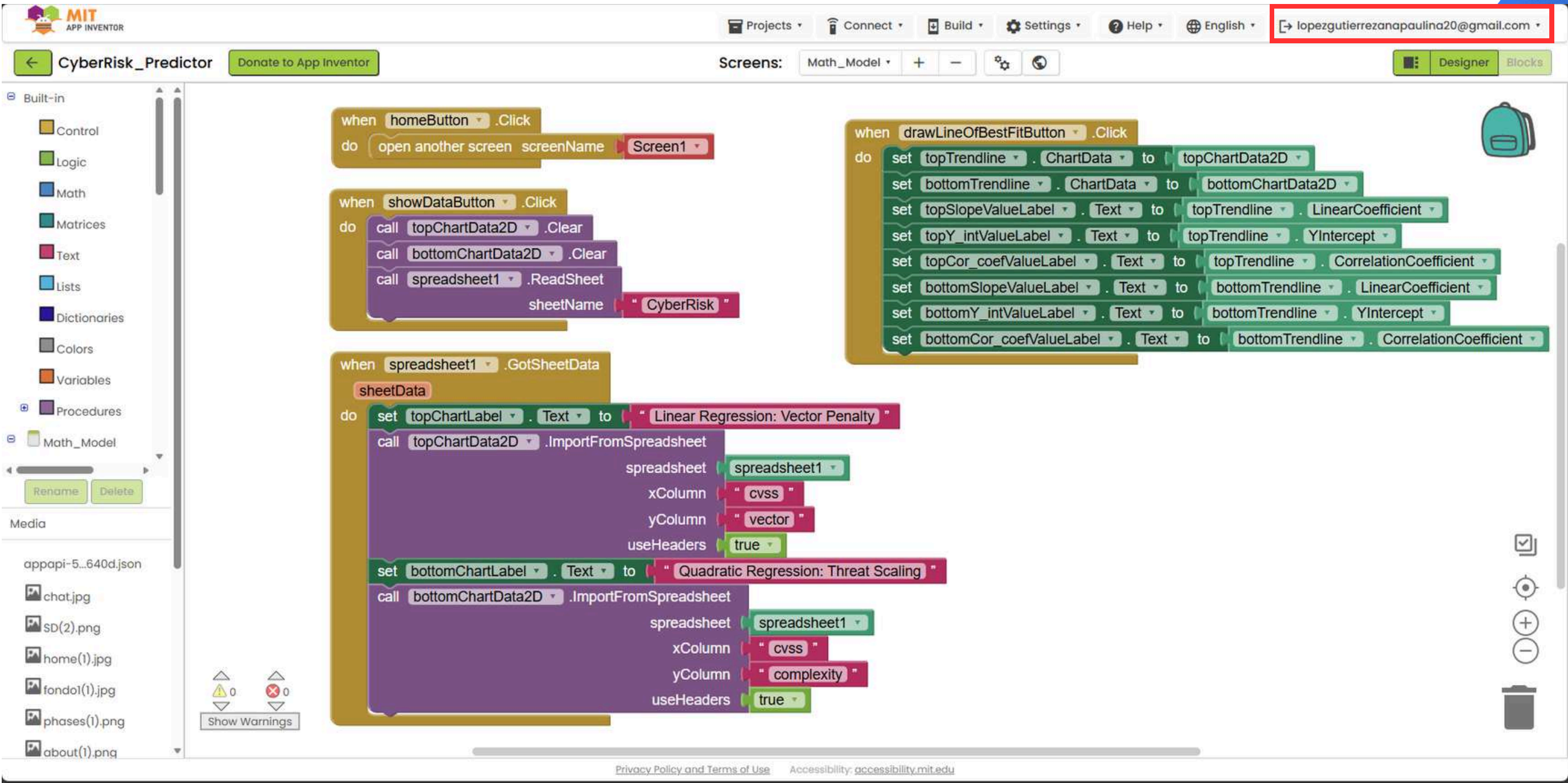
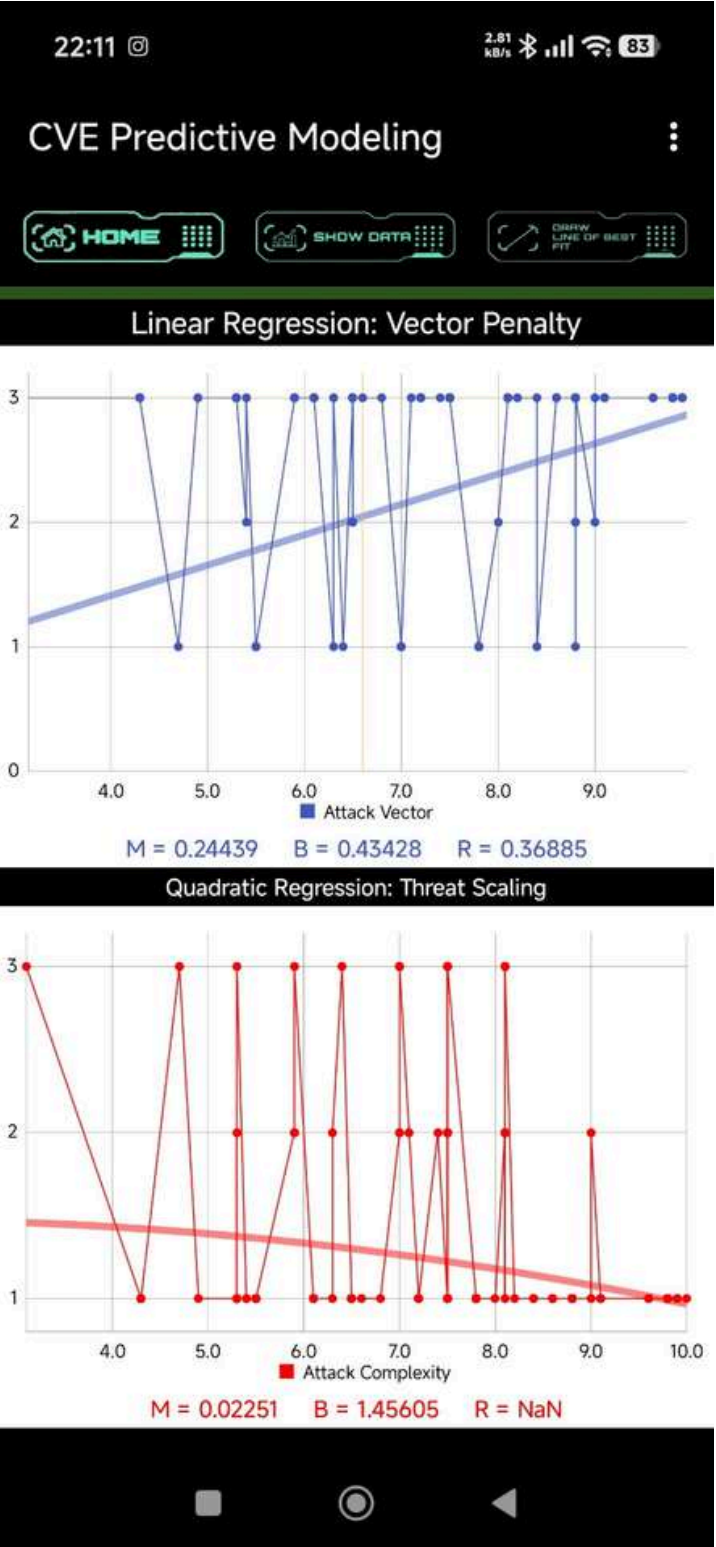


Interface

SCREEN

RESULT

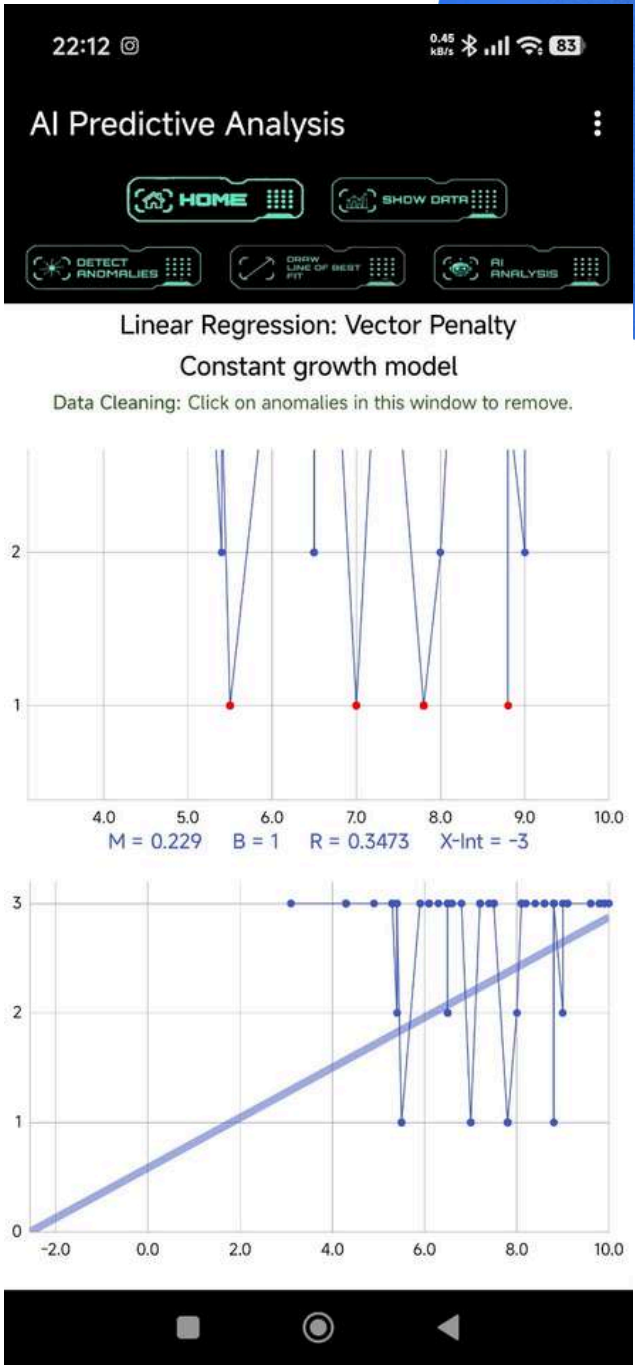
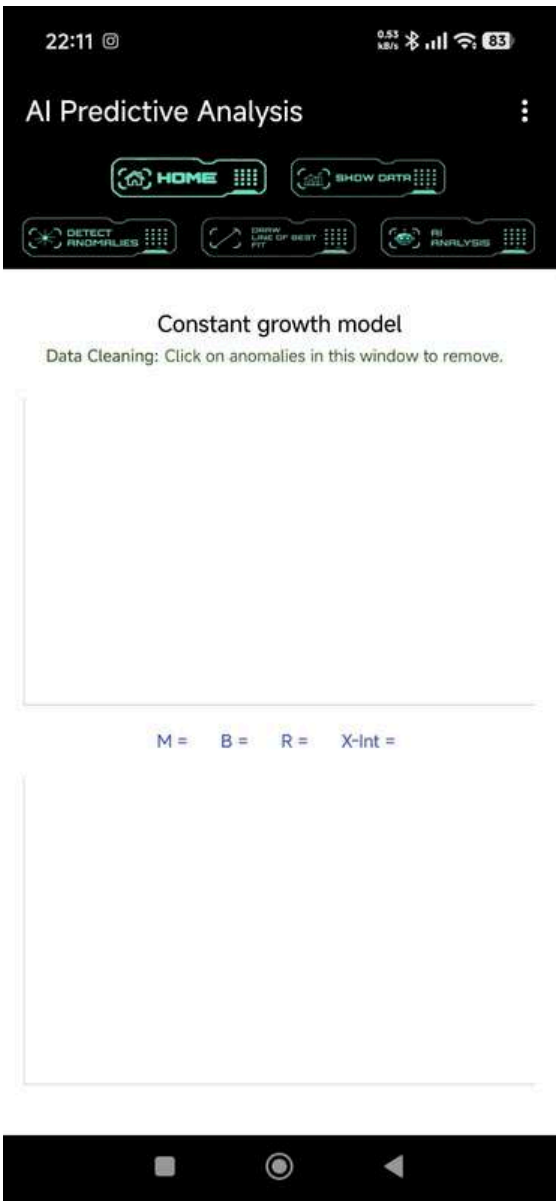
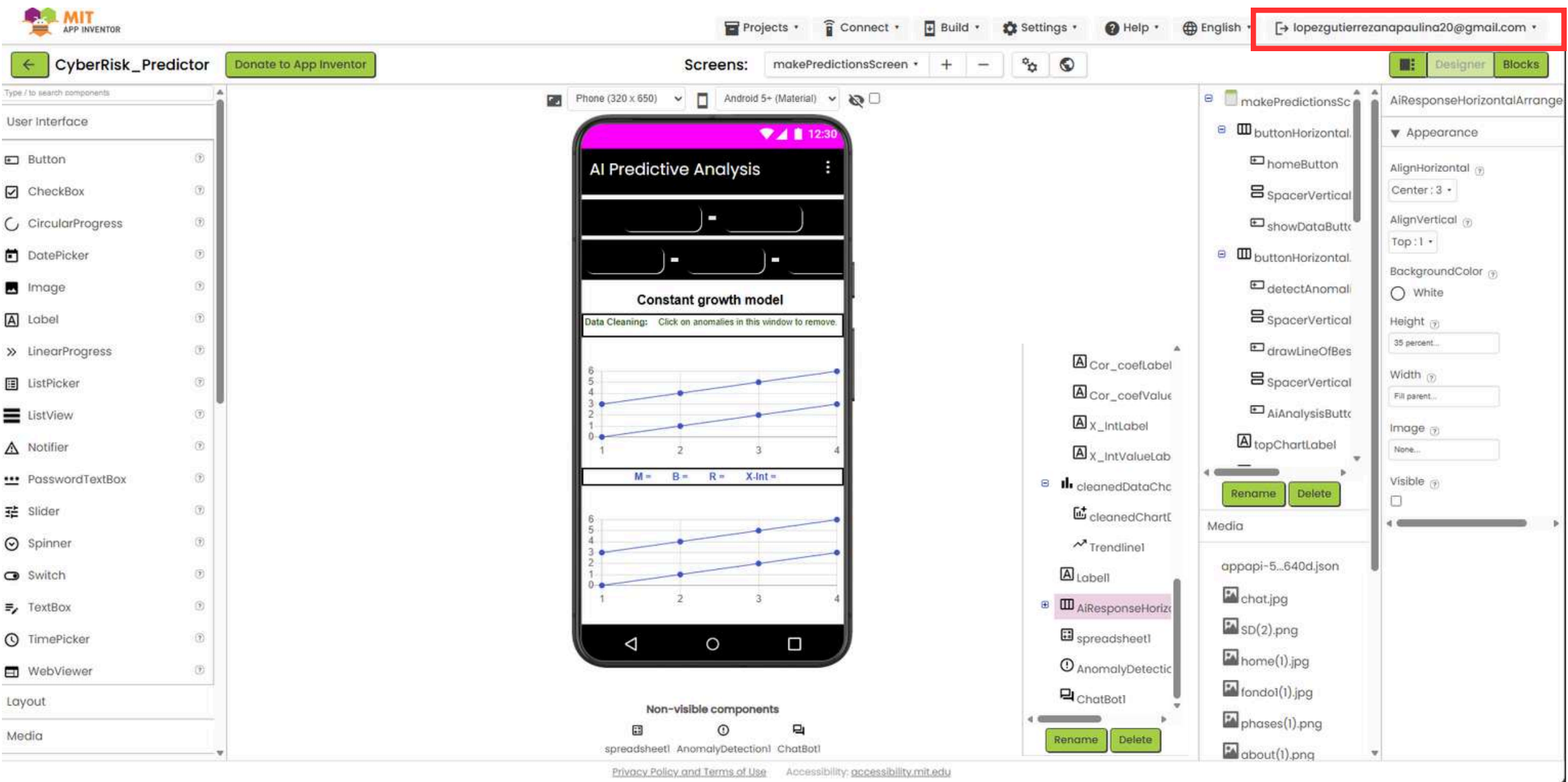




Interface

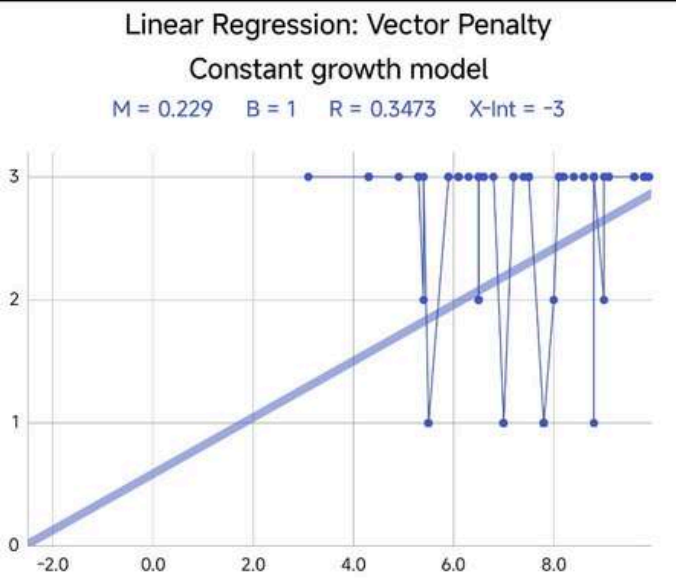
SCREEN

RESULT



Interface

RESULT

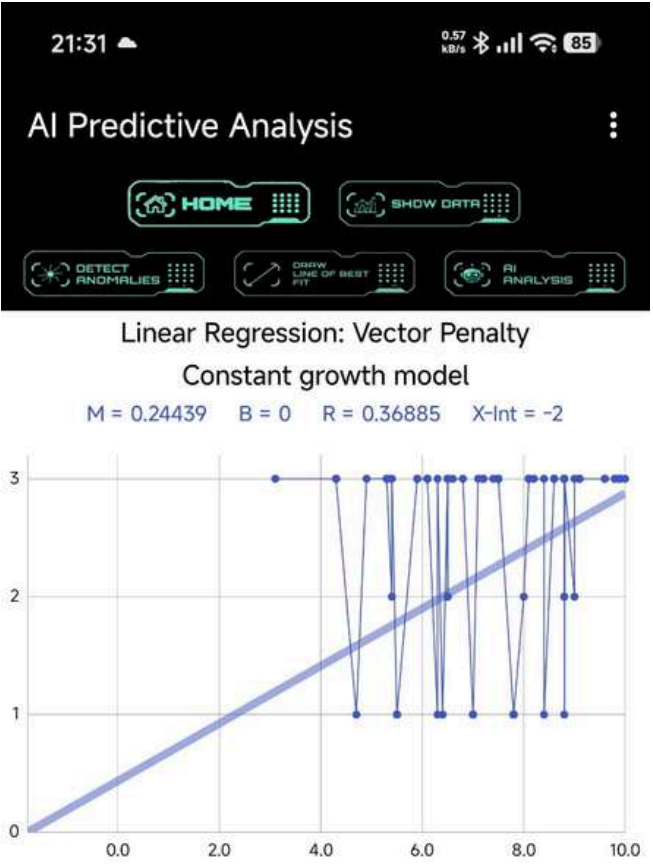


To estimate the CVSS risk score, we'd plug a 'network vector' value (assuming it represents a numerical input) into our linear equation:

$$\text{Estimated Score} = 0.229 * (\text{Network Vector}) + 0.58659$$

The result would be the predicted CVSS risk score for a critical vulnerability with that specific network vector.

****Implications:**** This trend suggests that network vectors play a modest role in determining CVSS scores, indicating other factors like exploit complexity and impact might be more influential. This highlights the need to prioritize robust security controls across all aspects of infrastructure, not solely focusing on network vulnerabilities.



To estimate the CVSS risk score for a critical vulnerability with a network vector, we'd use the equation of the line of best fit:

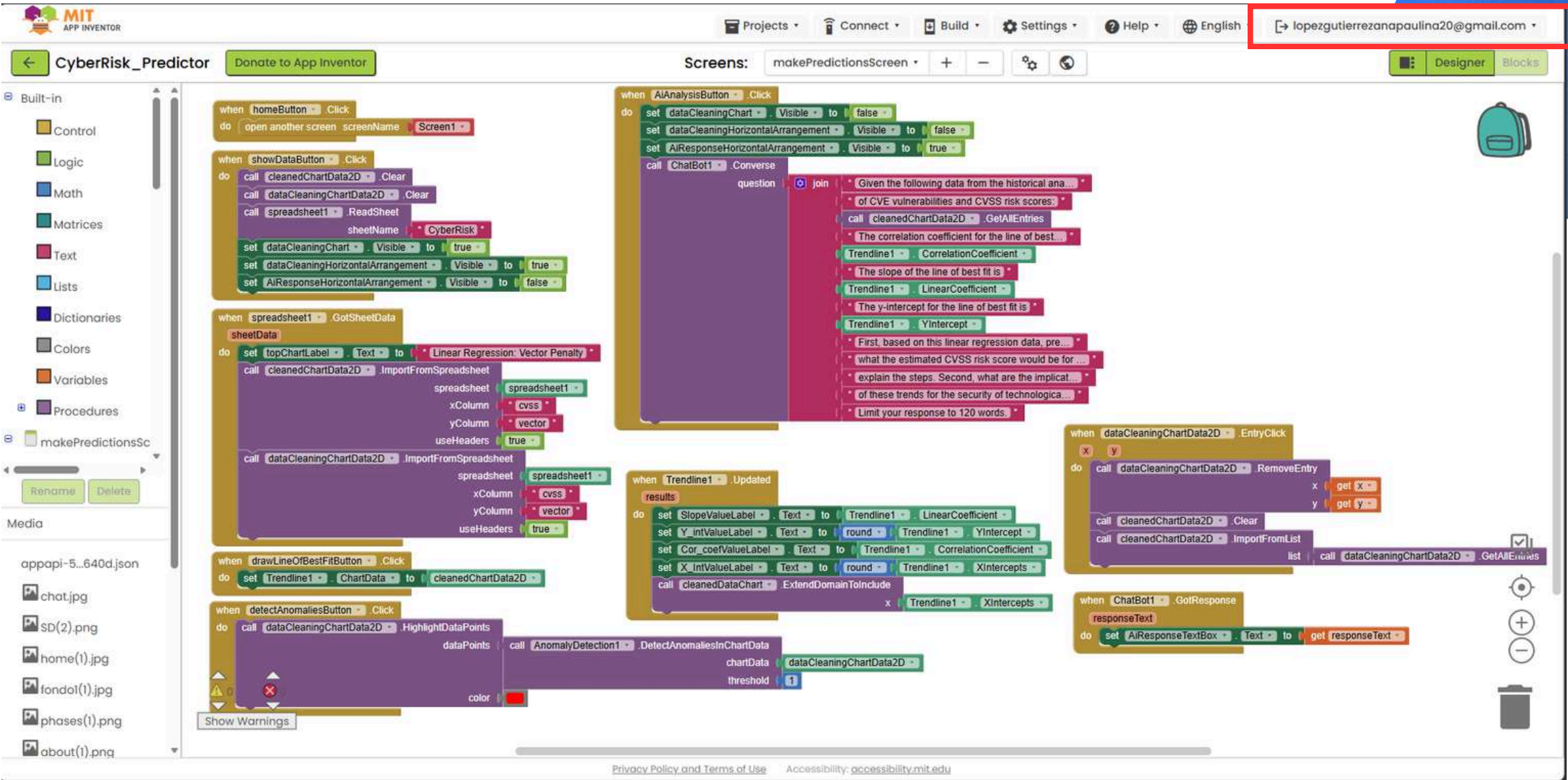
$$\text{**Score} = (\text{Slope} * \text{Network Vector}) + \text{Y-intercept**}$$

Plug in the provided slope and y-intercept, and the specific value of the network vector. The result will be the predicted CVSS risk score.

These trends suggest that network vectors have a moderate impact on CVSS scores, indicating security vulnerabilities exploiting network weaknesses pose a significant threat to technological infrastructure.

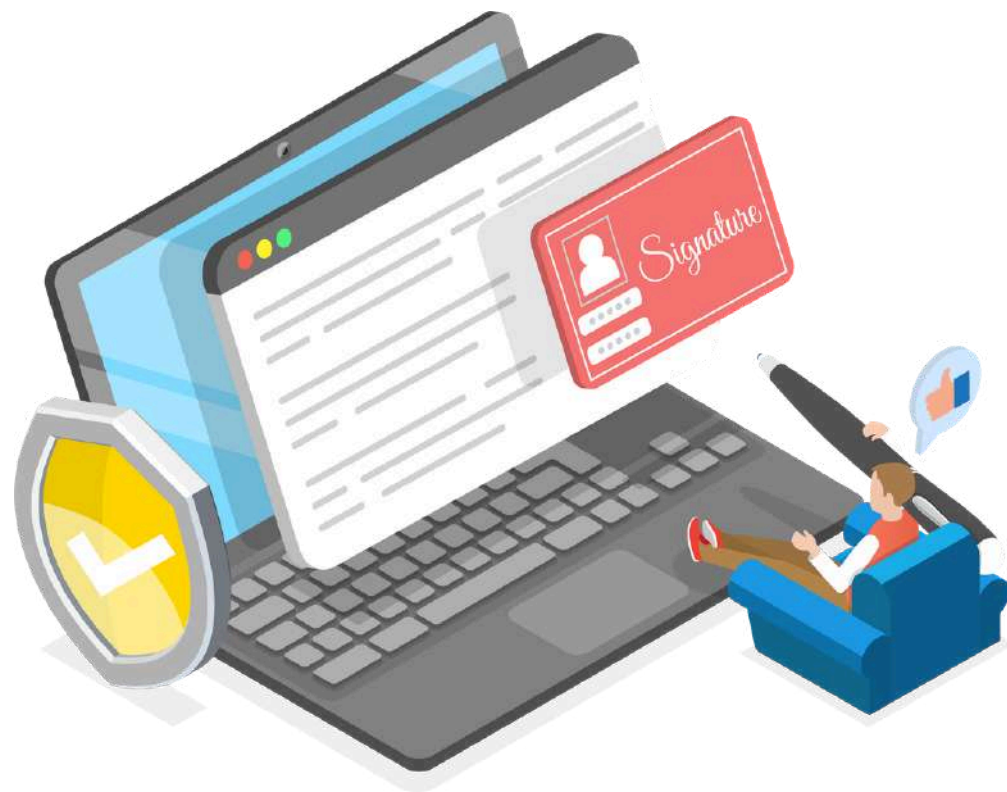


BLOCK PROGRAMMING



Conclusions

The CVE Database Explorer project successfully applied predictive mathematical models to analyze cybersecurity risks. By converting text categories into numbers, the application was able to calculate linear and polynomial regressions to understand how an attack's complexity affects its overall severity (CVSS score). Furthermore, the interactive data cleaning module showed that removing empty rows and invalid records from the original dataset is essential. This preprocessing step guaranteed that the app's mathematical trendlines were accurate and reliable.



Beyond graphing data, the project successfully integrated Artificial Intelligence in real-time. By feeding live mathematical results (such as the slope, y-intercept, and correlation coefficient) directly into a structured AI prompt, the app transformed from a simple data viewer into a smart diagnostic tool capable of providing practical cybersecurity recommendations. Ultimately, this project brought together database management, mobile app design, and data analysis. The final application not only meets its current goals but also creates a solid foundation for building more advanced threat intelligence systems in the future.